

I – The starting point

An internship with Bartek long (long) time ago

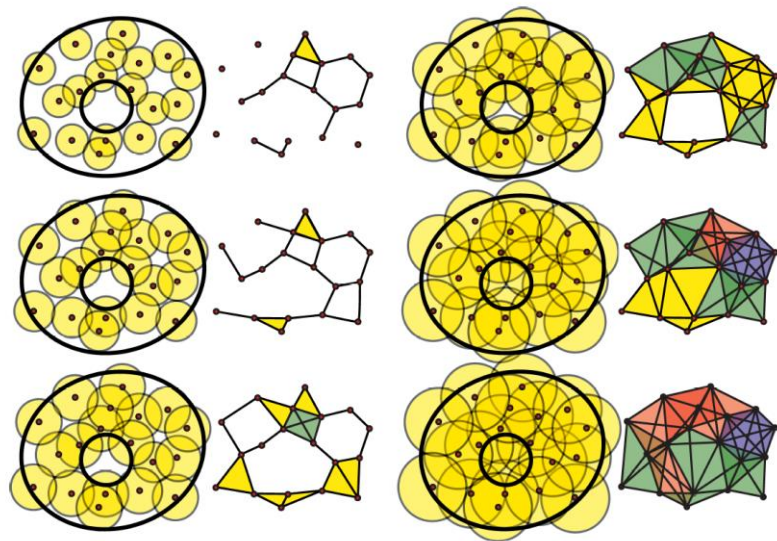
Un classificateur non-supervisé utilisant les
complexes simpliciaux
(avec une application à la *stylométrie*).

LOUIS HAUSEUX
Encadré par BARTŁOMIEJ BŁASZCZYSZYN
juillet 2017

Avant-propos

- On Topological Data Analysis
- Simplicial complexes and homology → recover topology of a manifold ?

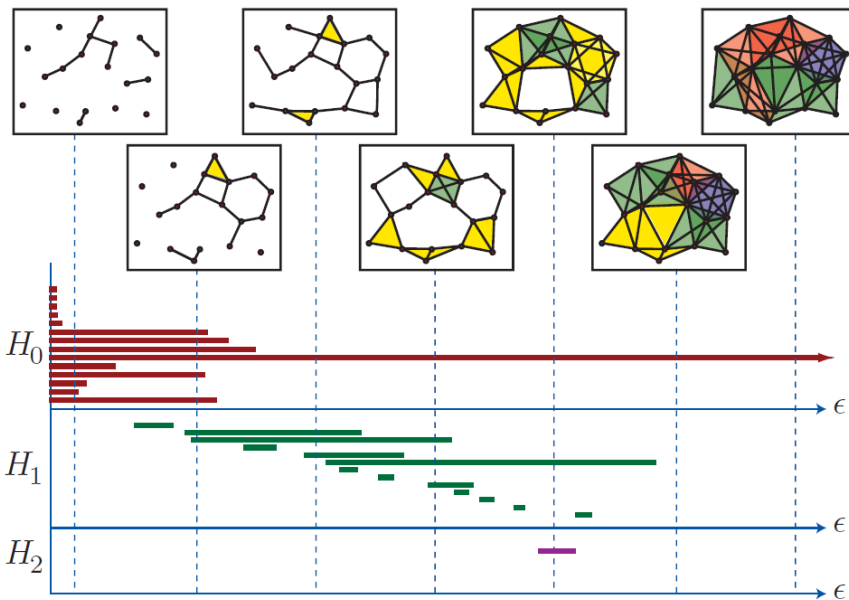
I – The starting point



Topological Data Analysis

- Make a radius r increase
- Look at the **homological structures** (generalisation of connect. comp., holes, etc.)

I – The starting point



Persistent Homology

- Make a radius r increase
- Look at the **homological structures** (generalisation of connect. comp., holes, etc.)

I – The starting point

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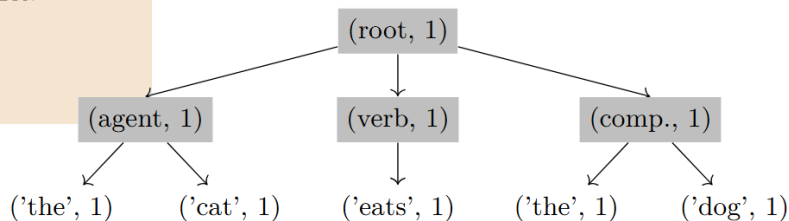
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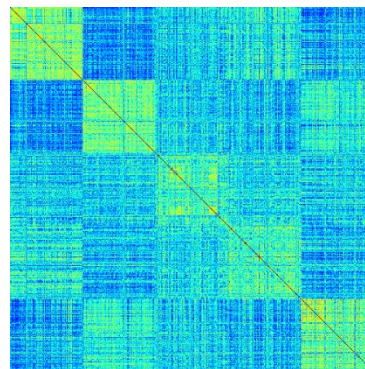
Avant-propos

Analogous ideas applied on a dataset:

- With **5** authors (journalists)
- **100** texts/author
- Goal: recover in a unsupervised way

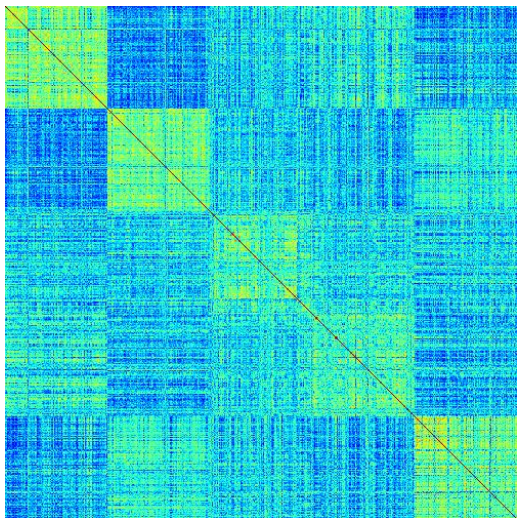


The **Gram-similarity matrix** →



I – The starting point

The **Gram**-similarity matrix



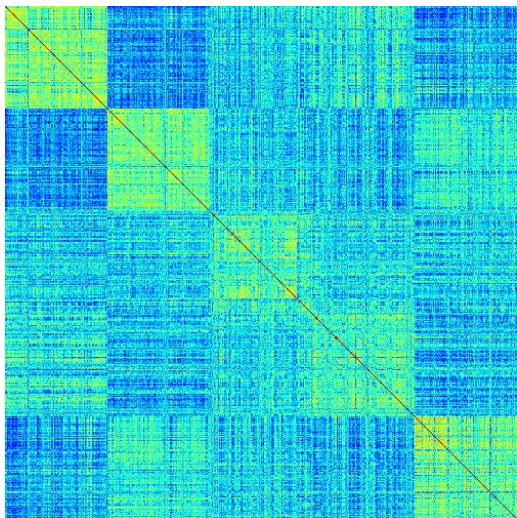
« Persistent Analysis »



- Make a radius r increase
- Observe '**simplexes-connected**' components (e.g. standard conn. comp. ; 'triangle-connected' comp. ; 'tetrahedron-connect.' &c.)

I – The starting point

The **Gram**-similarity matrix



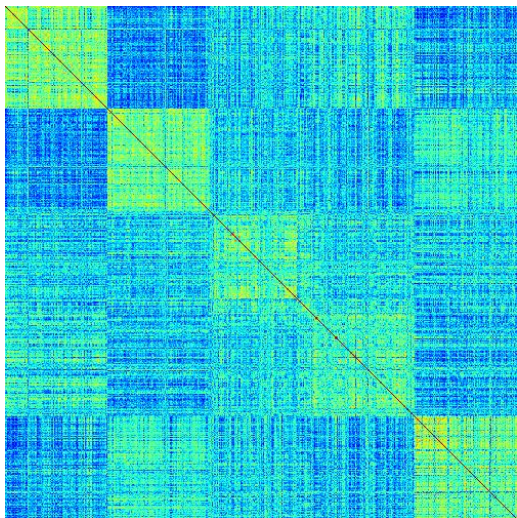
« Persistent Analysis »

- Make a radius r increase
- Observe '**simplexes-connected**' components (e.g. standard conn. comp. ; 'triangle-connected' comp. ; 'tetrahedron-connect.' &c.)

Already existed : **Clique percolation**
(on Vietoris-Rips complex)

I – The starting point

The **Gram**-similarity matrix



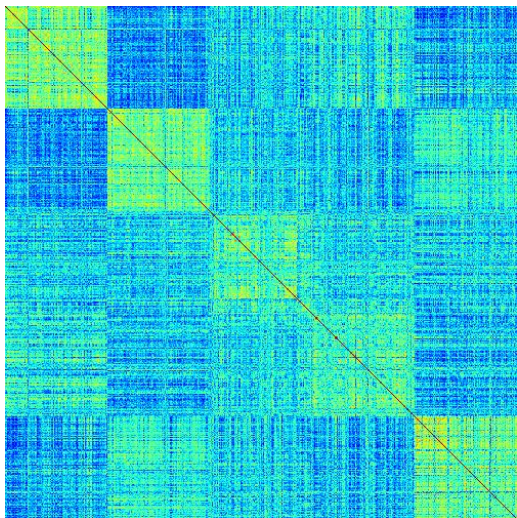
« Persistent Analysis »

- Make a radius r increase
- Observe '**simplexes-connected**' components (e.g. standard conn. comp. ; 'triangle-connected' comp. ; 'tetrahedron-connect.' &c.)

➔ It gave very good results !
(compared to K -means ; and even semi-superv. Gaussian mixtures models ; etc.)

I – The starting point

The **Gram**-similarity matrix



« Persistent Analysis »

- Make a radius r increase
- Observe 'simplexes-connected' components (e.g. standard conn. comp. ; 'triangle-connected' comp. ; 'tetrahedron-connect.' &c.)



➔ Why did it work so well ?



Inria Paris – MATHNET/DYOGENE seminar
2025-03-24

(Hyper-)graphs, percolation and clustering performances.

With

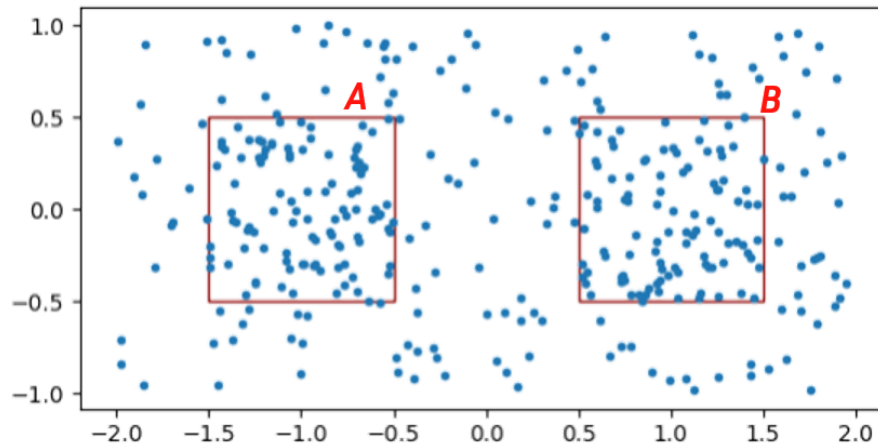
Konstantin Avrachenkov (NEO team) &
Josiane Zerubia (AYANA team)

Inria Sophia Antipolis

OVERVIEW

- I – Introduction: Starting point**
- II – Single-Linkage**
- III – The model: Statistician point of view**
- IV – The heart phenomenon: Percolation**
- V – Benefits of Hypergraphs**
- VI – Results: K -polyhedra $\gg$ HDBSCAN**
- VII – Collaboration with Nahuel**

II – Persistent Analysis on Geometric graphs

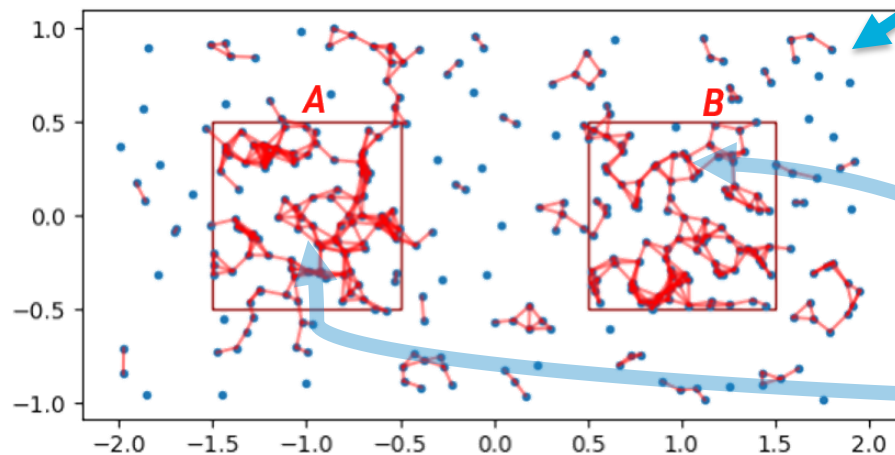


Old but efficient:
– Single-Linkage

Recent and state-of-the-art:
– (H)DBSCAN

II – Persistent Analysis on Geometric graphs

Geometric graph with radius
 $r \approx 0.14$

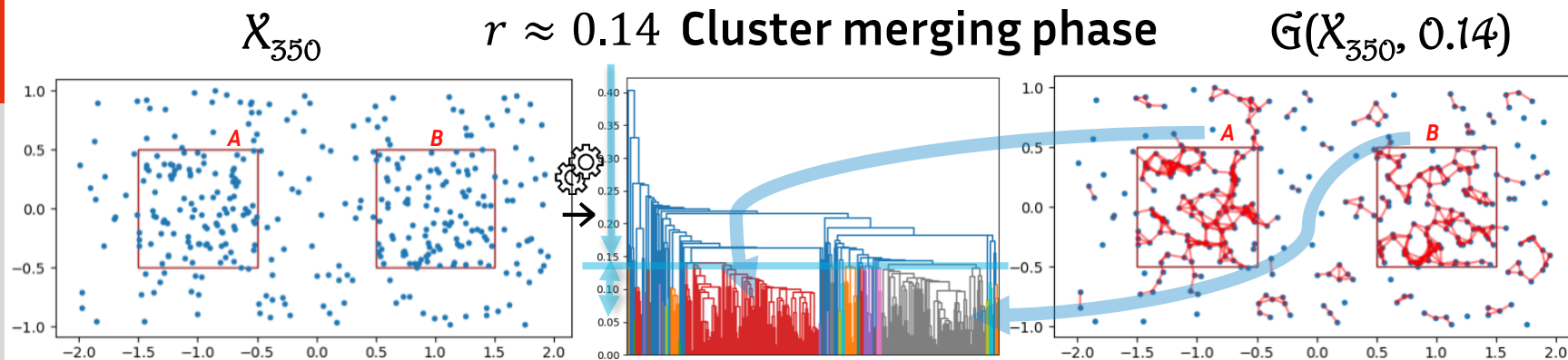


We almost recover the
High-Density Cluster

A
and **B**

II – Persistent Analysis on Geometric graphs

We get a hierarchical clustering



II – Geometric graphs ~ Minimum Spanning Tree

Theorem:

Sets of nodes of connect. comp. of $G(X,r)$ = Sets of nodes of *MST* pruned at r



→ It is sufficient to work with **Minimum Spanning Tree**

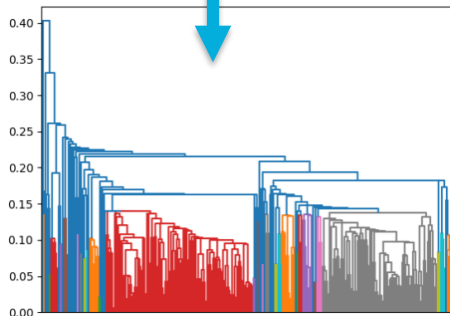
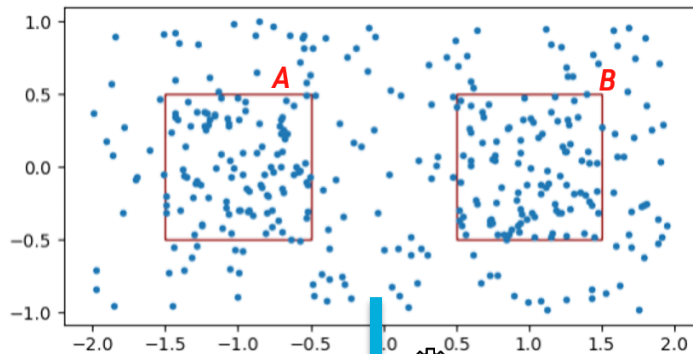
II – Geometric graphs ~ Minimum Spanning Tree

Theorem:

Sets of nodes of connect. comp. of $G(X,r)$ = Sets of nodes of *MST* pruned at r

- ➔ It is sufficient to work with **Minimum Spanning Tree**
- ➔ It is what **Single-Linkage** does

II – Single-Linkage



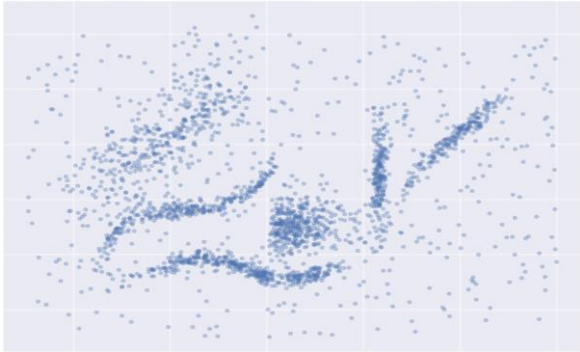
Single-Linkage:

- Begins with trivial clustering (n points, n clusters)
- At each step, merge the two closest clusters :

$$d_{\text{Clust}}(C, C') = \min_{x \in C, x' \in C'} \|x - x'\|$$

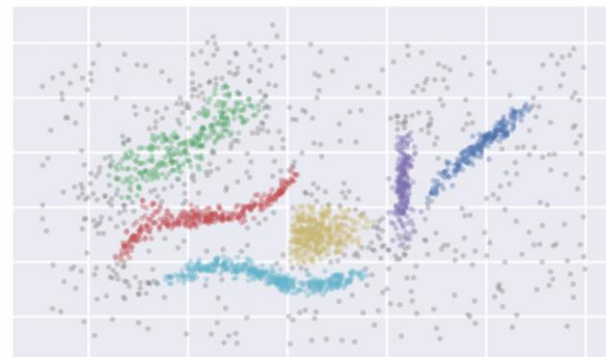
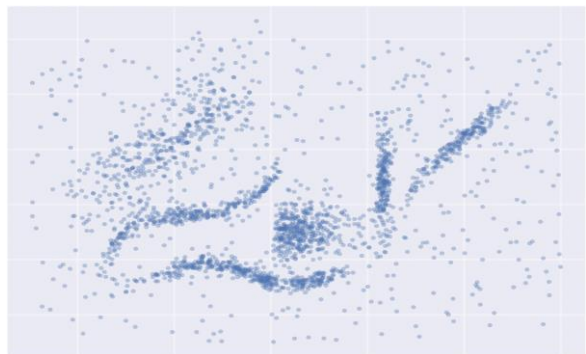
III – Standard stat. Model for clustering

Clustering



III – Standard stat. Model for clustering

Clustering



Hierarchical Density-
Based SCAN Algorithm *

* McInnes and Healy “Accelerated hierarchical density based clustering” (2017)

III – Model for clustering



What is the underlying structure (= the clusters) ?



→ The density f of point-generation



Suppose that all points are drawn IID w.r.t. to f

III – Model for clustering



What is the underlying structure (= the clusters) ?



→ The **density f** of point-generation



Goal: identify the « **High-Density Clusters** »*

*

J. Hartigan. *Clustering Algorithms* (1975)

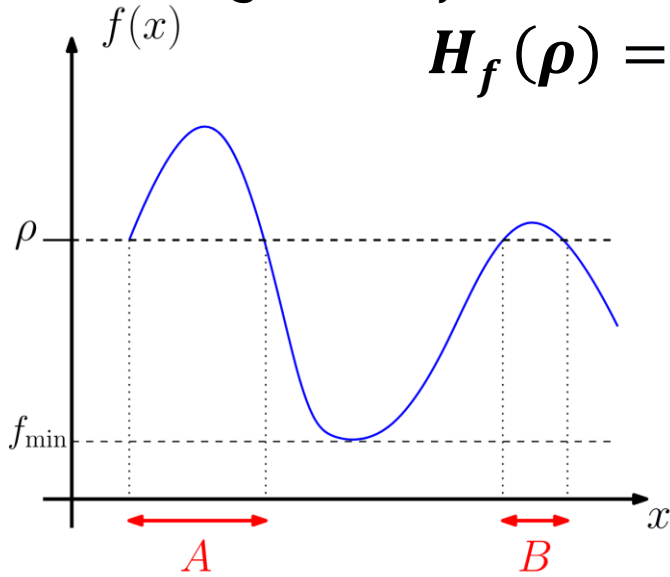
III – Model for clustering

High density clusters

High-Density Clusters $H_f(\rho)$ of level ρ : the connected components of:

$$H_f(\rho) = \{x \in \mathbb{R}^d \mid f(x) \geq \rho\} \subseteq \mathbb{R}^d$$

Important !



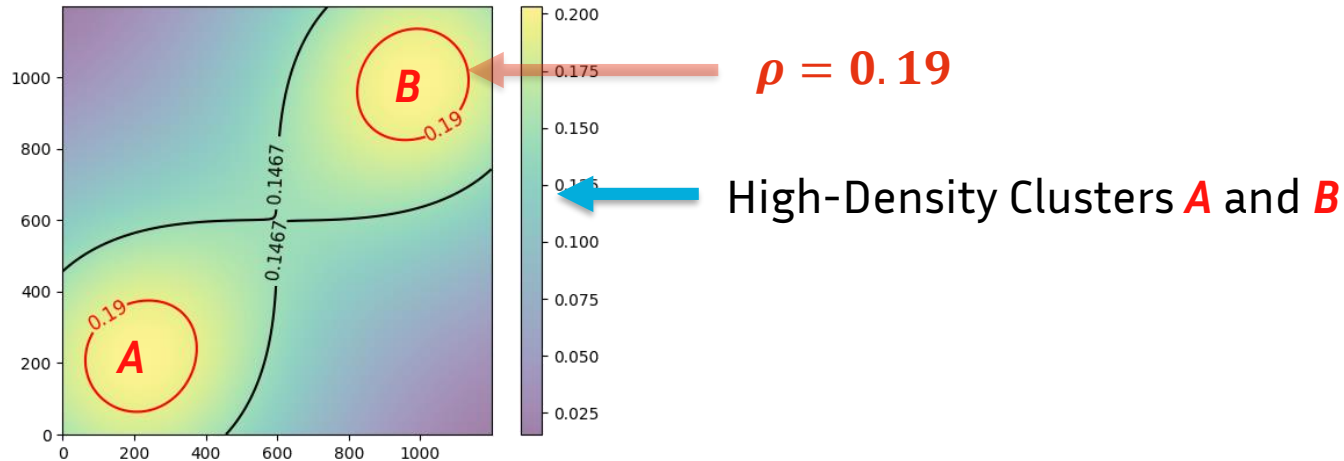
High-Density Clusters **A** and **B**

III – Model for clustering

High density clusters

High-Density Clusters $H_f(\rho)$ of level ρ : the connected components of:

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III – The model... and the problem

First solution: Computing an estimator $\hat{f}$ of f

$$H_f(\rho) \approx H_{\hat{f}}(\rho)$$



Problem: discretizing space is costly in $d = \dim(\mathbb{R}^d)$

III – Single-Linkage = 1-NN level set



1-Nearest Neighbor density estimator:

$$L_{\rho}^1 = \bigcup_{x \in \mathcal{X}_n} B(x, r) \quad \leftarrow \quad \rho = \frac{K}{n} \frac{1}{\omega_d} \left(\frac{1}{r}\right)^d$$

1-NN High-density



Clusterings of Single-Linkage

III – Single-Linkage: 3 points of view

- Geometric graphs → **percolation** phenomenon
- *1-NN High-density clusters* → statistical model
- Minimum Spanning Tree → efficient algorithms

III – HDBSCAN: a descendent of Single-Linkage



1) K -Robust Single-Linkage

Use consistency of **K -Nearest Neighbor density estimator**:

→ A node *must* have K neighbors to appear in geometric graphs

III – HDBSCAN: a descendent of Single-Linkage



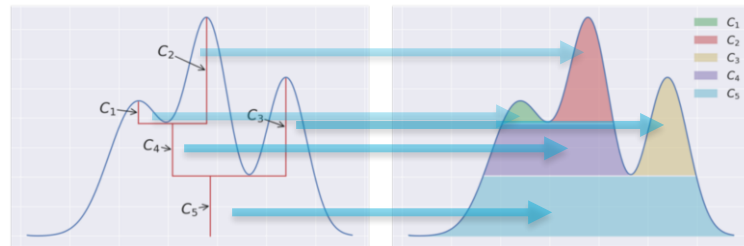
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Use consistency of **K -Nearest Neighbor density estimator**:

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2) Allow multi-scale clustering

$$E(C) = \int_C (f(x) - \rho) dx. \quad \text{Relative excess of mass:}$$



III – HDBSCAN: a descendent of Single-Linkage



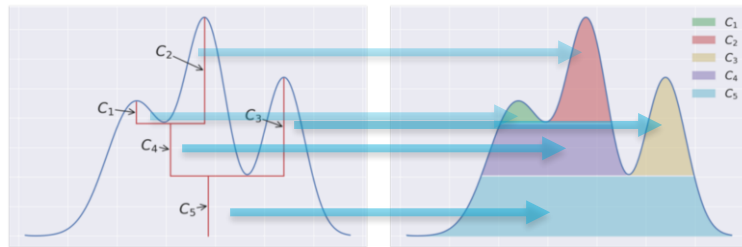
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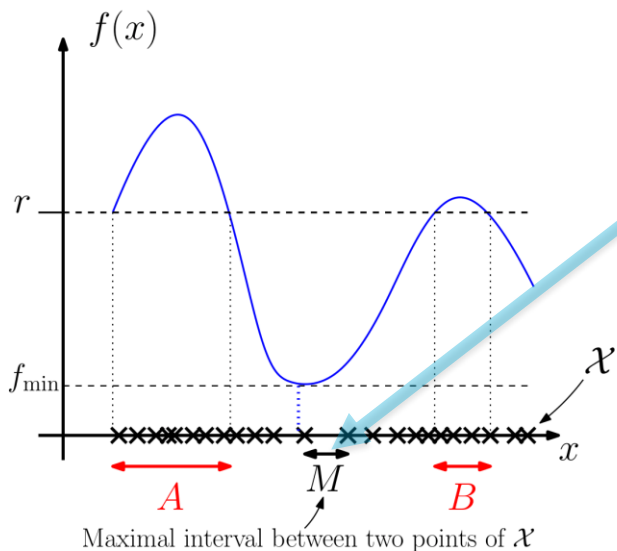
$$E(C) = \int_C (f(x) - \rho) dx. \quad \text{Relative excess of mass:}$$



3) Minimum size for clusters ← percolation phenomenon

IV – Why does it work in $\mathbb{R}$?

➔ Single-Linkage is **consistent** in $\mathbb{R}$!!!



The « cut » between the two clusters, containing **A** or **B**

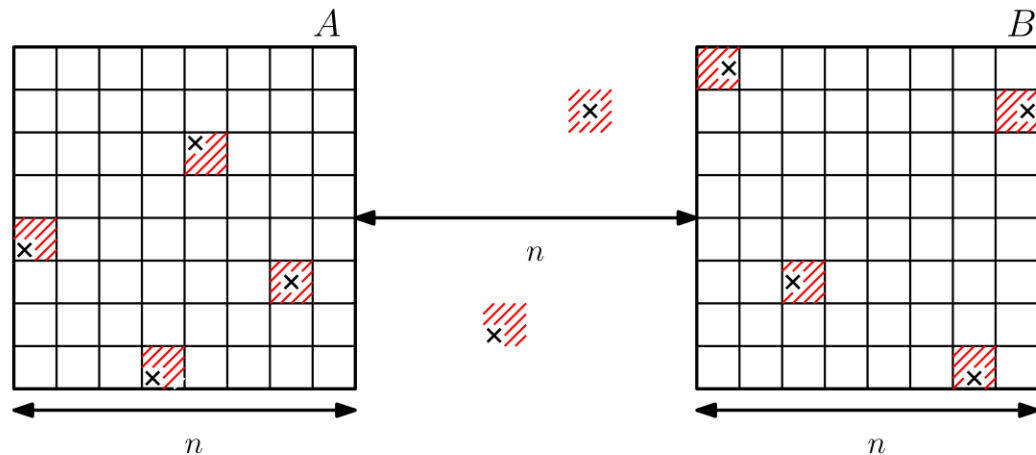
➔ But Single-Linkage is not **consistent** in $\mathbb{R}^d$ for $d \geq 2$...



IV – Why does it *not* work in $\mathbb{R}^d$? ($d \geq 2$)

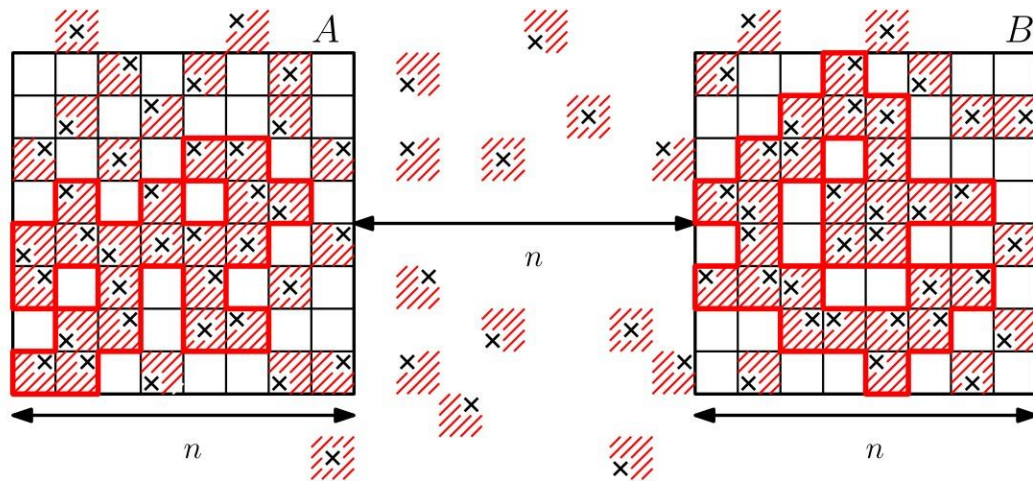
→ Single-Linkage is not consistent in $\mathbb{R}^d$ for $d \geq 2$... !!!

Discrete Site-Percolation on $\mathbb{Z}^2$



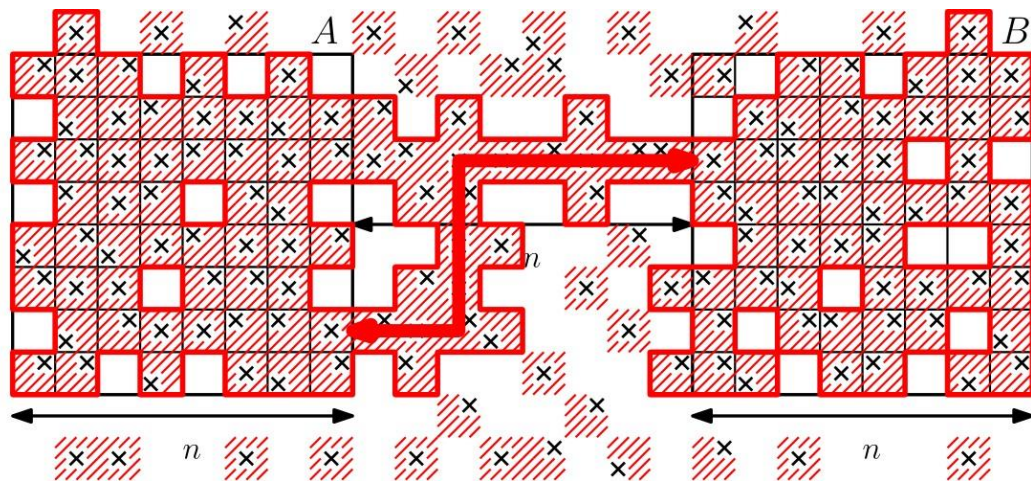
IV – Why does it *not* work in $\mathbb{R}^d$? ($d \geq 2$)

Percolation occurs on A and B



IV – Why does it *not* work in $\mathbb{R}^d$? ($d \geq 2$)

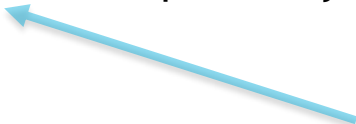
Percolation occurs everywhere...



➔ The two clusters merge before having recovered **A** and **B** entirely...

IV – Why does it *not* work in $\mathbb{R}^d$? ($d \geq 2$)

- Single-Linkage **works** in $\mathbb{R}^1$... because there is **no** percolation
- S.-L. **does not work** perfectly in $\mathbb{R}^d$... because **there is** percolation



Defeat on this battle... but war is not lost!

IV – Percolation: a conclusion

- Single-Linkage **works** in $\mathbb{R}^1$... because there is **no** percolation
- S.-L. **does not work** perfectly in $\mathbb{R}^d$... because **there is** percolation

Compare performance of clustering algo.

Studying the percolation phenomenon:

- A **fraction** of high-density clusters is recoverable...
- Which fraction? Define the **Percolation rate** *, **



*

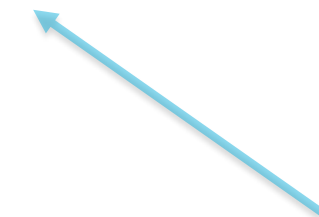
LH + KA + JZ, "Graph Based Approach for Galaxy Filament Extraction". Complex Networks (2023)

**

Work submitted to a journal

V - K -generalization of Single-Linkage

HDC of $1\text{-}NN$  Graph Connected Components



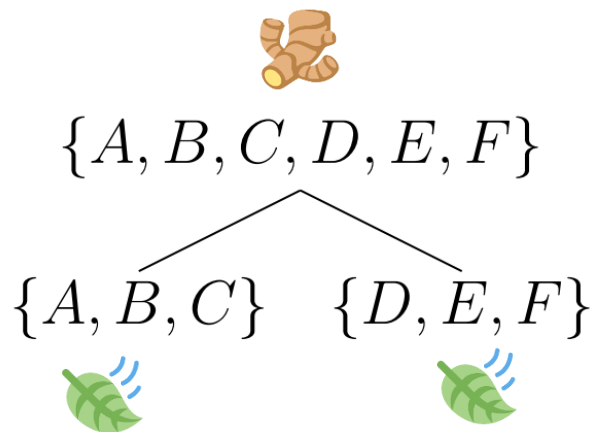
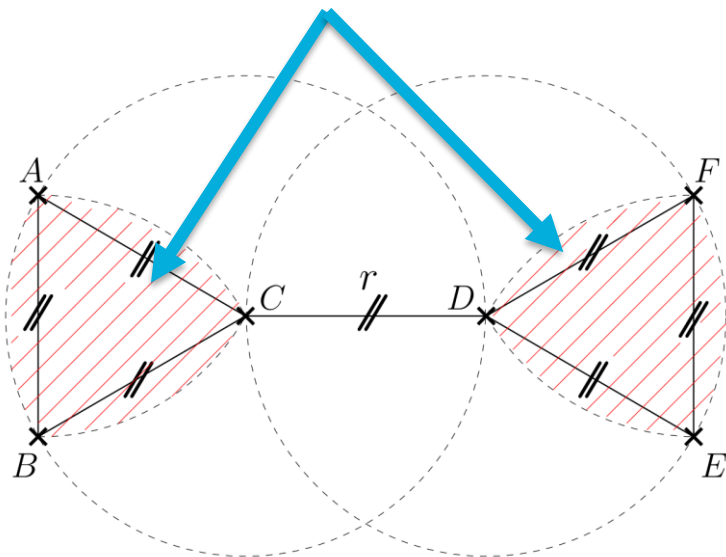
We would like $K\text{-}NN$ to gain in robustness

V – Hypergraphs... for K -high-density clusters

➔ Weakness of Robust Single-Linkage (or HDBSCAN)

The 3-high-density clusters of level r

Dendrogram of 3-NN density estimator



$V - K$ -high-density clusters $\sim$ Hypergraphs

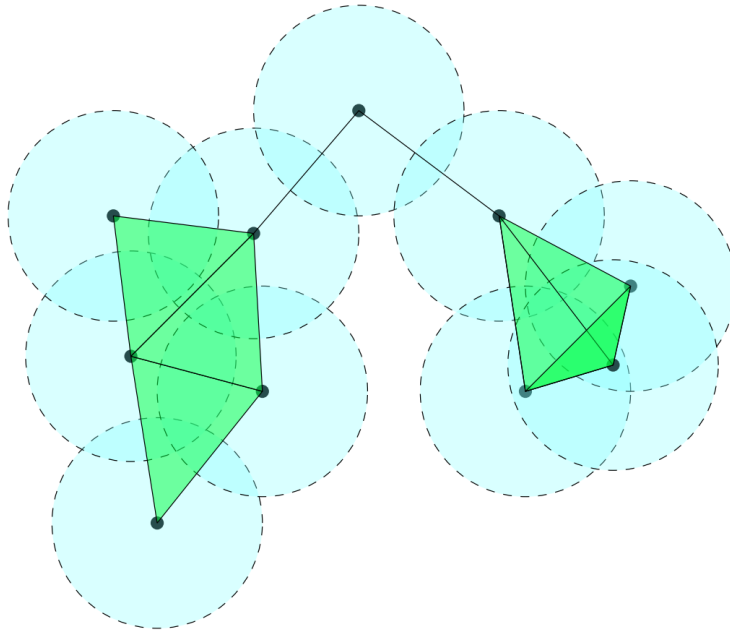
HDC of 1-NN $\longleftrightarrow$ Graph Connected Components



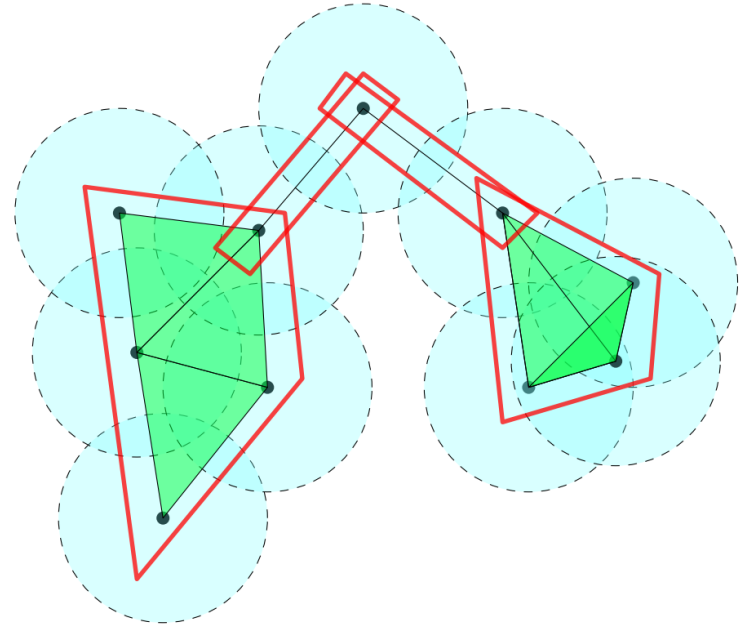
HDC of $K\text{-NN}$ $\longleftrightarrow$ Hypergraph Connect. Comp. *

* LH + KA + JZ, "Benefits of Hypergraphs for Density-Based Clustering". EUSIPCO (2024)

V – Hypergraphs, Polyhedra



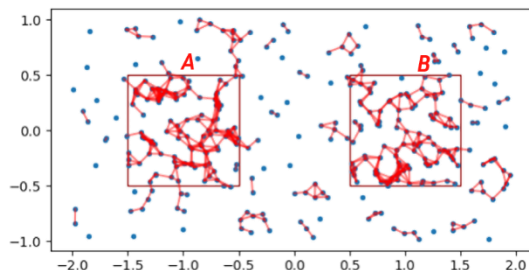
A hypergraph (Čech complex)



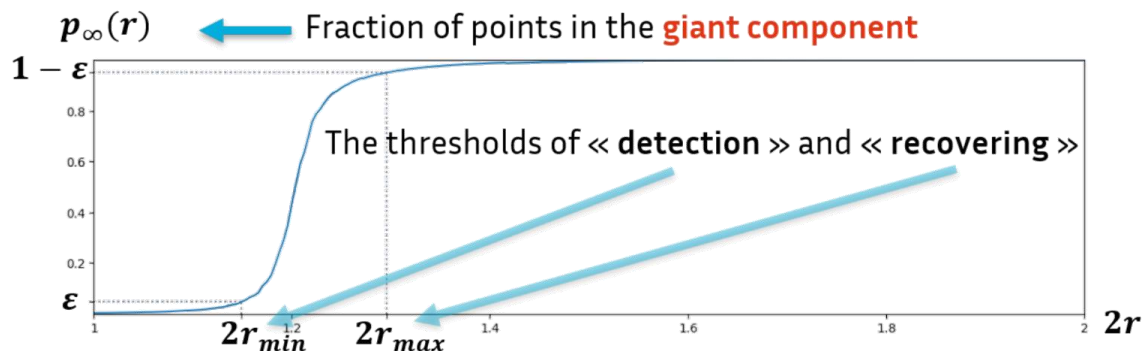
Polyhedra on a hypergraph

VI – Results: K -polyhedra $\gg$ HDBSCAN

1) In theory: the **percolation rate** of K -polyhedra *



The faster the percolation, the better the clustering

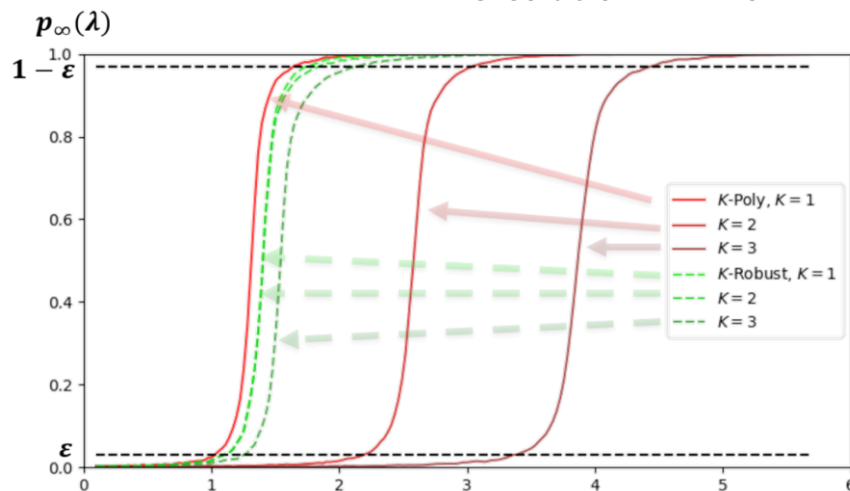


* LH + KA + JZ, "Hypergraphs, percolation, and hierarchical clustering". Complex Networks (2024)

VI – Results: K -polyhedra $\gg$ HDBSCAN

1) In theory: the **percolation rate** of K -polyhedra

Percolation in $\mathbb{R}^2$ for K -Robust S.-L. comp. and K -polyhedra



$\mathbb{R}^2$

Clustering	K -Robust Single-Linkage	K -polyhedra
$K=1$	$1.12/1.75 = \mathbf{0.64}$	$1.04/1.67 \approx 0.62$
$K=2$	$1.12/1.85 \approx 0.61$	$2.23/3.05 \approx \mathbf{0.73}$
$K=3$	$1.27/2.11 \approx 0.60$	$3.40/4.48 \approx \mathbf{0.76}$

$\mathbb{R}^3$

Clustering	K -Robust Single-Linkage	K -polyhedra
$K=1$	$0.42/1.05 = \mathbf{0.40}$	$0.41/1.05 \approx 0.39$
$K=2$	$0.42/1.26 \approx 0.33$	$1.26/2.36 \approx \mathbf{0.53}$
$K=3$	$0.53/1.60 \approx 0.33$	$2.37/3.81 \approx \mathbf{0.62}$

VI – Results: K -polyhedra $\gg$ HDBSCAN

2) In practice: the Olive oil dataset ^{*}, ^{**}

Macro-area	Region
South	1 – North Apulia
	2 – Calabria
	3 – South Apulia
	4 – Sicilia
Sardinia	5 – Inland Sardinia
	6 – Coast Sardinia
Centro-settentrionale	7 – East Liguria
	8 – West Liguria
	9 – Umbria

572 samples of oil composition
(vectors in $\mathbb{R}^8$)

Fatty acids: Palmitic, Palmitoleic,
Stearic, Oleic, Linoleic, Linolenic,
Arachidic, Eicosenoic



Can we recover the
geographic clusters given
only the fatty acids ?

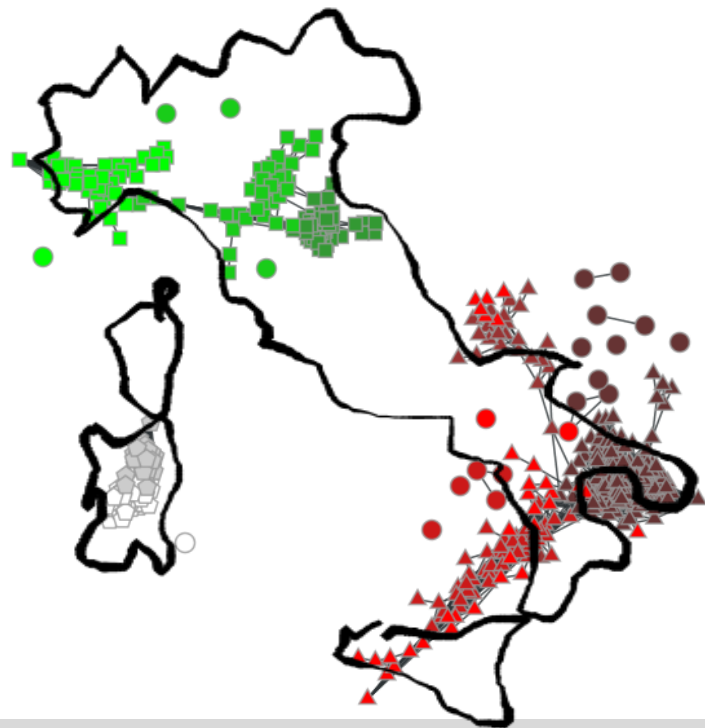
* Forina, Armanino, Lanteri & Tiscornia, "Classification of olive oils from their fatty acid composition" (1983)

** Rolle & Scoccola, "Stable and consistent density-based clustering" (2023)

VI – Results: K -polyhedra $\gg$ HDBSCAN

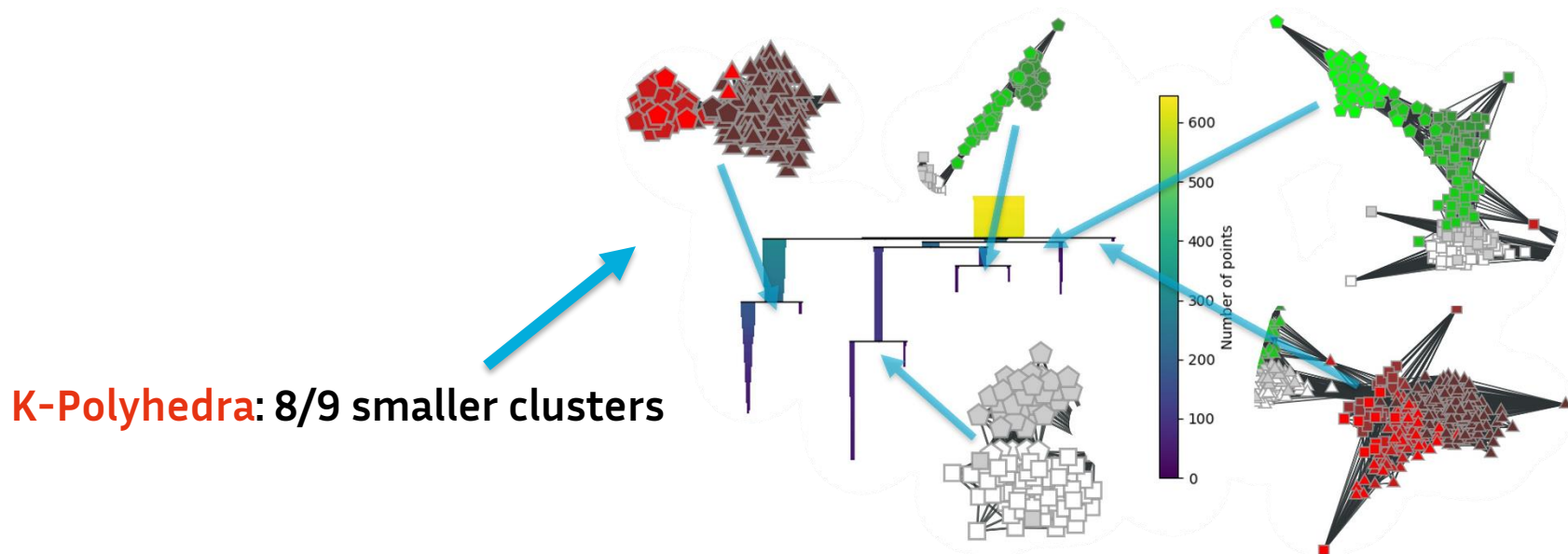
2) In practice: the Olive oil dataset

HDBSCAN: Only the 3 macro-areas



VI – Results: K -polyhedra $\gg$ HDBSCAN

2) In practice: the Olive oil dataset



VI – Results: K -polyhedra $\gg$ HDBSCAN

2) In practice: the Olive oil dataset

Confusion Matrix: Persistable * vs 2-polyhedra **

Pers./Ours	1	2	3	4	5	6	7	8	9	Miss.
N. Apulia	12/24	0/0	0/0	-	0/0	0/0	0/0	0/0	0/0	13/1
Calabria	0/0	7/25	1/2	-	0/0	0/0	0/0	0/0	0/0	48/29
S. Apulia	0/0	0/0	100/145	-	0/0	0/0	0/0	0/0	0/0	106/61
Sicilia	3/6	0/1	0/2	-	0/0	0/0	0/0	0/0	0/0	33/27
Inl. Sard.	0/0	0/0	0/0	-	51/62	0/0	0/0	0/0	0/0	14/3
Coast S.	0/0	0/0	0/0	-	0/2	19/29	0/0	0/0	0/0	14/2
E. Ligur.	0/0	0/0	0/0	-	0/0	0/0	14/32	1/2	0/1	35/15
W. Ligur.	0/0	0/0	0/0	-	0/0	0/0	0/0	29/33	0/0	21/17
Umbria	0/0	0/0	0/0	-	0/0	0/0	0/0	0/0	42/46	9/5



412/572 points clustered

relative accuracy:
396/412 $\approx$ 96.1%

*

**

Rolle & Scoccola, "Stable and consistent density-based clustering" (2023)

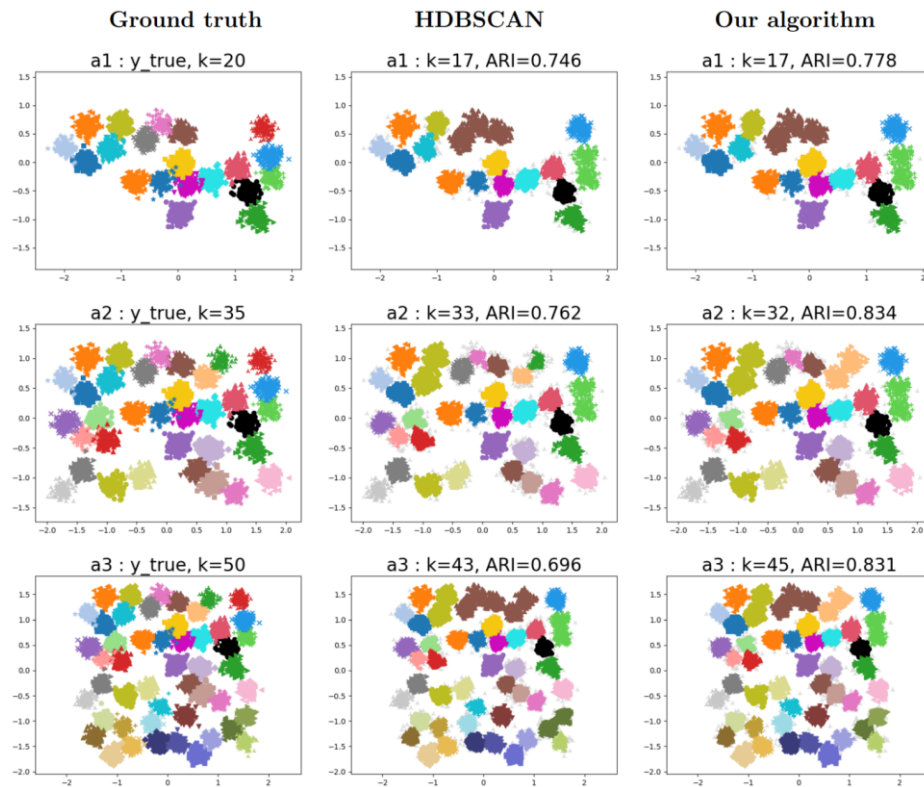
Work to be submitted to SIAM Journal on Mathematics of Data Science

Inria

VI – Results: K -polyhedra $\gg$ HDBSCAN

2) In practice:
the SIPU benchmark $*$, $**$

- The existence of **ground truth**
- The diversity of the datasets:
 - On the size
(from $n = 373$ to $n = 105600$)
 - On the shapes of clusters
(not only Gaussian)
 - On the number k of clusters (from $k = 2$ to $k = 100$)
 - On the difficulty of clustering
- The dimension $d = 2$ of the data space



*

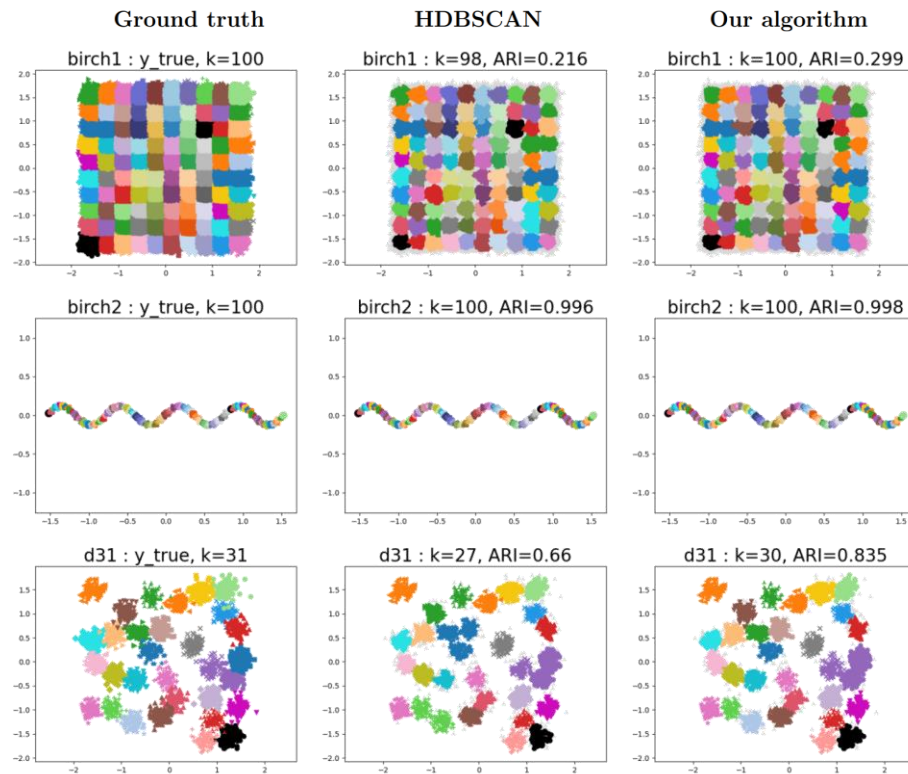
**

Gagolewski, "A framework for benchmarking clustering algorithms" (2022)
Work to be submitted to SIAM Journal on Mathematics of Data Science

VI – Results: K -polyhedra $\gg$ HDBSCAN

2) In practice: the SPU benchmark

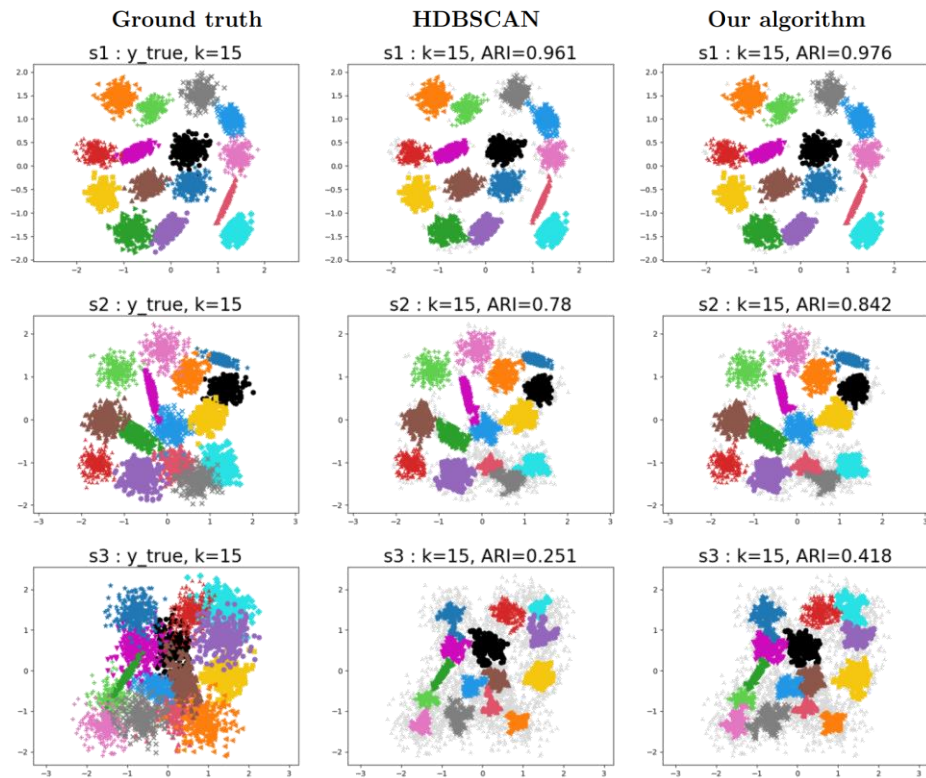
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VI – Results: K -polyhedra $\gg$ HDBSCAN

2) In practice: the SIPU benchmark

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VI – Results: K -polyhedra $\gg$ HDBSCAN

2) In practice: the SIPU benchmark

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(not only Gaussian)
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- The dimension $d = 2$ of the data space

Dataset	n	Ground truth	HDBSCAN		Our method	
		k	k	ARI	k	ARI
a1	3000	20	17	0.745	17	0.778
a2	5250	35	33	0.762	32	0.834
a3	7500	50	43	0.696	45	0.831
aggregation	788	7	5	0.809	5	0.809
birch1	100000	100	98	0.216	100	0.299
birch2	100000	100	100	0.996	100	0.998
d31	3100	31	27	0.66	30	0.835
jain	373	2	3	0.935	3	0.935
s1	5000	15	15	0.961	15	0.976
s2	5000	15	15	0.78	15	0.842
s3	5000	15	15	0.251	15	0.418
s4	5000	15	15	0.258	15	0.284
spiral	312	3	3	1	3	1
unbalance	6500	8	8	1.000	8	1.000
worms_2	105600	35	15	0.055	33	0.076

VI – Results: K -polyhedra $\gg$ HDBSCAN

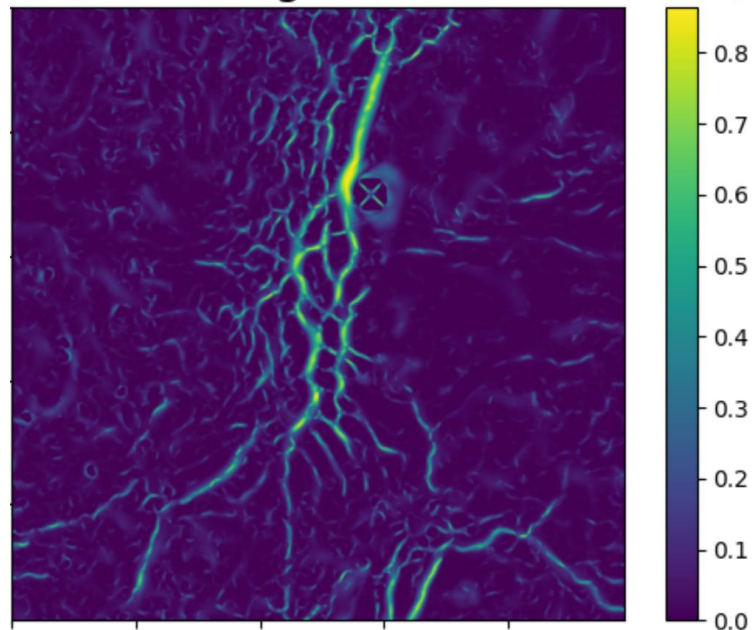
2) In practice: generalized Frangi filter *



Response
pixelwise



Frangi filter



*

Work submitted for the next EUSIPCO conference (2025)

VI – Results: K -polyhedra $\gg$ HDBSCAN

2) In **practice**: generalized Frangi filter

(Hyper-)graph extraction

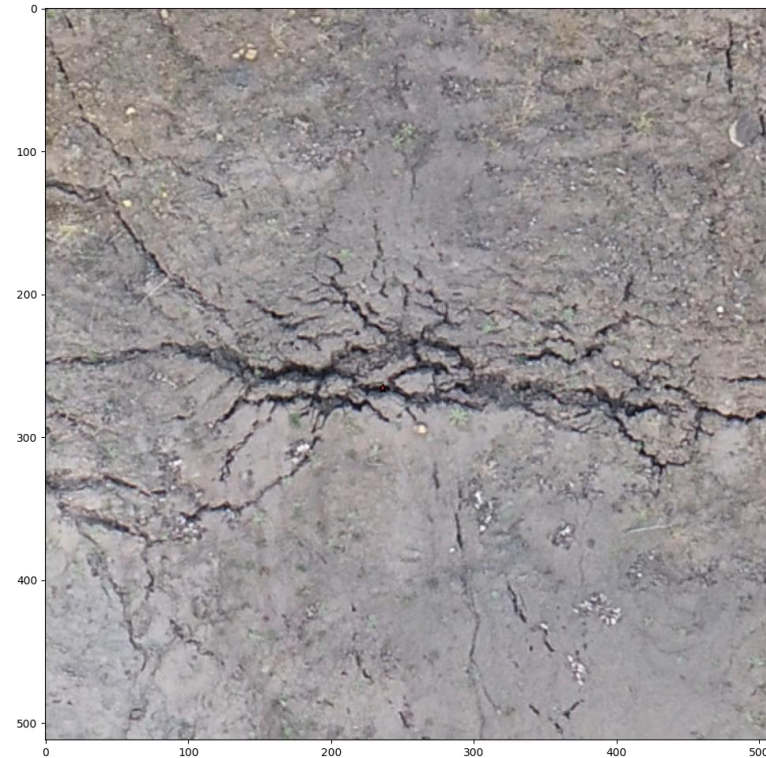


Response on
couples of pixels



VI – Results:

2) In practice:
generalized Frangi filter



VII – Collaboration with Nahuel

1) First model: the geometry of the distribution gives the communities

2) **What about if the linking connection creates the communities ?**

SANKARAMAN & BACCELLI
(ACM-SIAM SODA 2018)



Doi:10.1137/1.9781611975031.142

→ Points i.i.d. generated

→ Spatial Connections $f_{in}(\cdot)$ and $f_{out}(\cdot)$

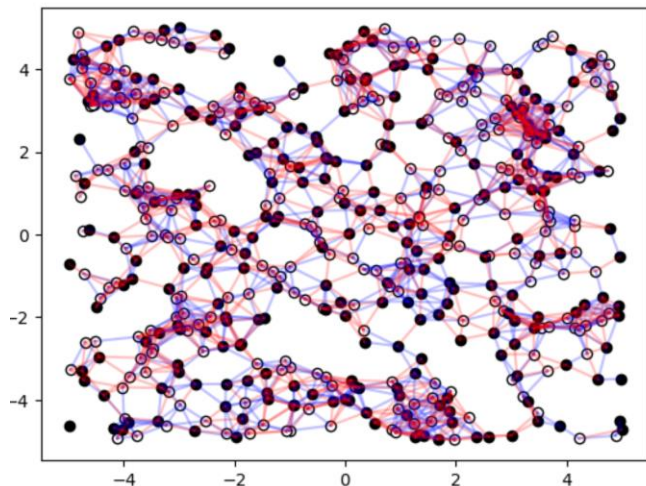
→  Application : haplotypes

Community Detection on Euclidean Random Graphs*

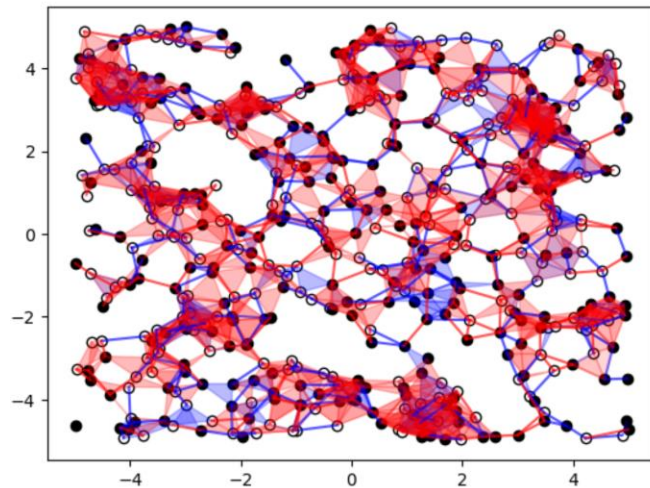
Abishek Sankararaman,[†] François Baccelli[‡]

VII – Collaboration with Nahuel

- 1) First model: the geometry of the distribution gives the communities
- 2) What about if the linking connection creates the communities ?



Ferromagn. &
anti-ferro.
interactions
→
Higher-order



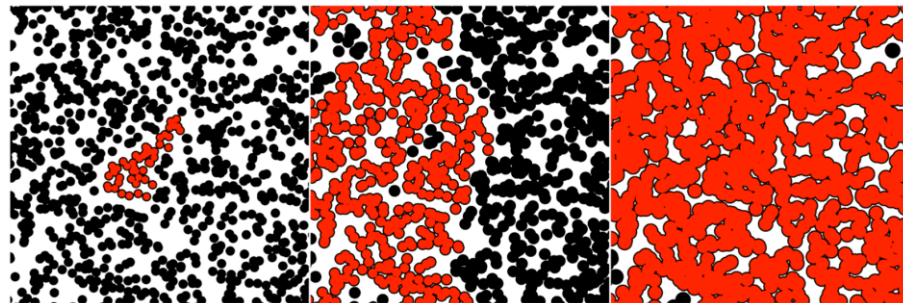
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IV – The Percolation *

Model : the $x_1, \dots, x_n$ IID plotted uniformly on $[0 ; 1]^2$.

→ Poisson Point Process $\lambda \leftarrow 1$



Geometric Graphs / greatest component
 $r < r_c$ $r = r_c$ $r > r_c$

Percolation = giant component

→ Very fast phenomenon

Two sub-cases : there exists a critical value $r_c \approx 1.2$:

- If $r < r_c$, no great component (of size $\Theta(\log(n))$)
- If $r > r_c$, **one giant component** (of size $\Theta(n)$)

*

M. Penrose. *Random Geometric Graphs* (2003)